

Functional Requirements

Project Name

Smart Lender – Loan Eligibility Prediction System

Introduction

Functional requirements describe the core functionalities that the Smart Lender system must perform. These requirements define how the system interacts with users, processes data, and generates loan eligibility predictions using machine learning models.

Functional Requirements

1. User Interface

- The system shall provide a simple and user-friendly web interface.
 - Users shall be able to navigate between the Home, Prediction, and Result pages.
 - The interface shall be developed using HTML, CSS, and Flask.
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2. Applicant Information Input

The system shall allow users to enter the following applicant details:

- Gender
 - Married Status
 - Number of Dependents
 - Education
 - Self Employed Status
 - Applicant Income
 - Co-applicant Income
 - Loan Amount
 - Loan Amount Term
 - Credit History
 - Property Area
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3. Data Validation

The system shall validate all user inputs before processing.

- Mandatory fields must not be empty.

- Numeric values must contain only numbers.
 - Invalid entries should display appropriate error messages.
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4. Data Preprocessing

The system shall preprocess the input data before prediction by performing:

- Missing value handling
 - Categorical value encoding
 - Feature scaling
 - Data formatting compatible with the trained model
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5. Loan Prediction

The system shall load the trained machine learning model and predict whether the loan application is:

- **Approved**
- **Rejected**

based on the applicant's details.

6. Result Display

The prediction result shall be displayed immediately after submission.

The result page shall clearly indicate:

- Loan Approved
 - Loan Rejected
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7. Model Loading

The system shall load the saved machine learning model (saved_model.pkl) during application startup to ensure efficient predictions.

8. System Performance

The application shall generate predictions quickly with minimal response time.

9. Error Handling

The system shall handle invalid or incomplete inputs gracefully and prevent application crashes.

Functional Modules

The Smart Lender project consists of the following modules:

- Home Module
 - Prediction Module
 - Machine Learning Prediction Module
 - Result Module
 - Flask Backend Module
 - Model Loading Module
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Conclusion

The functional requirements define the essential operations of the Smart Lender system. These functionalities ensure that users can easily provide applicant information, receive accurate loan predictions, and interact with the application efficiently.